



Telefónica – AWS Infrastructure, SaaS Development & Global Data Migration

The Telefónica Global Infrastructure Transformation Programme delivered a unified, cloud-native foundation across 46,000 end-user devices spanning 750 locations and multiple countries and time zones.

This initiative modernised core infrastructure, accelerated SaaS adoption, and enabled secure, scalable data migration to Amazon Web Services (AWS) — establishing a global digital platform for innovation and operational excellence.

Programme Overview

Telefónica's legacy infrastructure required modernisation to support rapid growth, distributed operations, and evolving customer demands.

The programme introduced a cloud-first architecture, leveraging AWS to deliver elasticity, resilience, and cost efficiency across the enterprise.

Key Deliverables

AWS Infrastructure Modernisation:

- Designed and deployed a multi-region AWS environment supporting compute, storage, and networking workloads.
- Implemented Infrastructure-as-Code (IaC) using Terraform and CloudFormation for repeatable, compliant provisioning.
- Established automated monitoring, patching, and backup frameworks aligned to ISO 27001 and GDPR standards.

SaaS Development & Integration:

- Delivered new SaaS applications for internal and customer-facing services, built on microservices and containerised architectures.
- Integrated identity, billing, and analytics modules with AWS Lambda, API Gateway, and DynamoDB.
- Enabled continuous integration and deployment (CI/CD) pipelines using AWS CodePipeline and GitHub Actions.

Data Migration & Cloud Storage:

- Migrated structured and unstructured data from on-premises systems to AWS S3, RDS, and Redshift.
- Ensured data integrity through checksum validation, encryption at rest and in transit, and automated reconciliation.
- Delivered global data-replication strategy to maintain performance across time zones.

End-User Estate Transformation:

- Supported 46,000 devices across 750 locations, introducing secure connectivity, identity federation, and endpoint management.
- Implemented unified access policies and device compliance via AWS Workspaces and Intune integration.
- Delivered early-life support and user adoption campaigns to ensure seamless transition.

Operational Governance & Service Management:

- Established ITSM integration with ServiceNow for incident, change, and configuration management.
- Authored service designs, operational manuals, and runbooks for cloud operations.
- Delivered training and knowledge-transfer sessions to internal teams across Europe, Latin America, and Asia.



Outcomes

- Global AWS infrastructure supporting 46,000 users across 750 locations
- Modern SaaS ecosystem enabling faster product delivery and innovation
- Secure, compliant data migration with improved performance and resilience
- Unified operational model across multiple time zones
- Reduced infrastructure costs and improved scalability



Strategic Impact

The Telefónica AWS Transformation Programme established a global digital backbone — connecting people, platforms, and data across continents.

It delivered a future-ready environment capable of supporting continuous innovation, operational agility, and sustainable growth.